

Rigidity Factors in Unimolecular Reactions

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Unimolecular reactions with loose activated complexes, i.e. simple bond fission and the reverse bond formation processes, are conveniently characterized by phase space theory (PST), which accounts for the maximum attraction potential between the fragments, and by additional rigidity factors which are the result of the anisotropy of the potential. This article summarizes recent calculations of specific rigidity factors $f_{\text{rigid}}(E, J)$ and of thermal rigidity factors $f_{\text{rigid}}(T)$ for a series of different potentials such as, e.g., ion-dipole, dipole-dipole, or standard valence potentials. The calculations have been made by rigorous statistical adiabatic channel (SACM) in combination with classical trajectory (CT) calculations.

1. Introduction

Statistical rate theory provides the by far most widely used framework for characterizing unimolecular and related reaction processes. The central quantity of this treatment is the specific rate constant $k(E, J)$ of the unimolecular reaction step. At the energy E and the angular momentum J , $k(E, J)$ is given by

$$k(E, J) = W(E, J) \times (1/h \rho(E, J)) \quad (1.1)$$

where the second factor $1/h \rho(E, J)$ denotes the *one-channel rate constant* for reaction into a single specified product state and the first factor $W(E, J)$ is the *number of open reaction channels*, each of which is characterized by a one-channel rate constant.

It should be emphasized that the two factors have to be considered separately, because they concern different aspects of the problem. The *one-channel rate constant* includes all details of intramolecular vibrational energy redistribution. Harmonic oscillator models of $\rho(E, J)$, such as used in most practical applications, provide an approximate estimate. However, anharmonicity and intramolecular couplings have to be included as well, see e.g. [1–5]. In the energy range of a dissociation continuum, the boundaries of the phase space to be included in $\rho(E, J)$ require careful investigation of the dissociation dynamics. The *number of open channels* concerns a different detail of the reaction. Under the conditions of adiabatic dynamics [6–8], the maxima of adiabatic channel potential curves have to be determined. Tunneling contributions may have to be included in $W(E, J)$. Under the conditions of nonadiabatic dynamics, branching of the dynamics into several adiabatic channel potentials enters $W(E, J)$ as well.

In the present work, we focus attention on the *number of open channels* only. Considering $W(E, J)$, the reaction may be characterized by two different situations, either by “rigid” or by “loose” activated complexes. Provided that normal mode frequencies along the reaction path can be calculated in the vicinity of rigid activated complex barriers, state counting of standard RRKM theory directly leads to $W(E, J)$, see e.g. [9]. In the following we only consider the opposite case of *loose activated complexes* where angular momentum conservation requires particular attention. This

case, to a first approximation, is treated by phase space theory (PST) in which numbers of states of conserved oscillators are convoluted with numbers of rovibrational states of separated fragments, see e.g. [10–14]. It is recommended to elaborate PST with the true maximum attraction potential between the dissociation fragments, i.e. not to use unrealistic van der Waals potentials in PST. The anisotropy of the potential surface afterwards should be accounted for by *specific rigidity factors* $f_{\text{rigid}}(E, J)$ [15] to be multiplied with $W(E, J)$ from phase space theory, i.e. by splitting $W(E, J)$ through

$$W(E, J) = f_{\text{rigid}}(E, J) W^{\text{PST}}(E, J) \quad (1.2)$$

In this way, the kinetic relevances of the attraction and of the anisotropy of the potential can be rationalized in the most transparent way.

Even without solving the problems of the one-channel rate constant $1/h \rho(E, J)$, understanding $W(E, J)$ opens the way to a series of kinetic observables. Thermal dissociation and combination reactions in the high pressure limit contain the thermal average only over $W(E, J)$. Apart from trivial factors, the rate constants are proportional to *activated complex partition functions*

$$Q^*(T) = \sum_{J=0}^{\infty} (2J+1) \int_{E_0(J)}^{\infty} W(E, J) \exp(-E/kT) dE/kT \quad (1.3)$$

with the J -dependent threshold energies $E_0(J)$.

In this case, *thermal rigidity factors* $f_{\text{rigid}}(T)$ [16] appear, defined by

$$f_{\text{rigid}}(T) = Q^*(T)/Q^{*\text{PST}}(T) = k/k^{\text{PST}} \quad (1.4)$$

Eq. (1.2) also enters directly the rate coefficients for complex-forming bimolecular reactions [18] in the low pressure limit, i.e.

$$k \propto \sum_{J=0}^{\infty} (2J+1) \int_{E_{\text{oa}}(J)}^{\infty} W_a(E, J) \times \left\{ \frac{W_b(E, J)}{W_a(E, J) + W_b(E, J)} \right\} \exp(-E/kT) dE/kT \quad (1.5)$$

where the index a corresponds to the activated complex at the entrance and b to that of the exit of the intermediate reaction adduct. Likewise, state-resolved cross sections are expressed, e.g., by

$$\sigma(v, j, E_i \rightarrow v', j', E'_i) = \frac{\pi}{(2j+1)k_{vj}^2} \sum_{J=0}^{\infty} (2J+1) W(E, J, v, j) \left\{ \frac{W(E, J, v', j')}{W(E, J, v, j) + W(E, J, v', j')} \right\} \quad (1.6)$$

with $k_{vj} = (2\mu E_i)^{1/2}/\hbar$ and channel-specific numbers of open channels $W(E, J, v, j)$ (or $W(E, J, v', j')$) for a process of the type $A + BC(v, j) \xrightarrow{E_i} ABC^* \xrightarrow{E'_i} A + BC(v', j')$ [19]. Only when $k(E, J)$ is measured directly or in competition to other processes, such as collisional energy transfer, $1/h\rho(E, J)$ besides $W(E, J)$ enters the observable. Falloff curves of thermal unimolecular reactions, pressure-dependent rate constants of complex-forming bimolecular reactions, or pressure dependent quantum-yields of photolysis processes, thus contain the complete expressions of $k(E, J)$.

Our method of approach is the statistical adiabatic channel model (SACM) [19, 20] in its rigorous form and combined with classical trajectory (CT) calculations such as elaborated in [6–8, 21–23]. By comparison of SACM and CT in the range of classical dynamics and by considering the relevant Massey parameters as adiabaticity parameters, it was shown that rigorous SACM provides an accurate representation of the dynamics for typical practical conditions. Only at superthermal energies the dynamics becomes “globally” nonadiabatic and the construction of adiabatic channel potential curves becomes useless. However, before *global* nonadiabaticity sets in, *local* nonadiabatic branching of the dynamics at avoided crossings of adiabatic channel potential curves may occur [24]. These are comparably rare when one dissociation fragment is an atom [24], but become very frequent when both dissociation fragments are diatomic or polyatomic [23]. In this case, except for very low channel quantum numbers, a CT treatment can hardly be avoided.

Our SACM and combined SACM/CT calculations provide an accurate representation of the unimolecular simple bond fission and the reverse radical association dynamics which even allows to include quantum effects in open electronic shell systems, see e.g. [21] and [25]. In this way, specific and thermal rigidity factors can be determined as a function of the most important parameters of the potential energy surface. Starting from rudimentary forms [17], we now follow this concept more systematically and rigorously to potentials of increasing complexity [6–8, 15, 22]. Because of the simple final expressions it should be particularly suited for experimental applications.

Before describing rigidity factors for special cases, our approach should be compared with alternative treatments by *variational transition state theory* (VTST) in *canonical* (CVTST) or *microcanonical* (μ VTST) form. In the present issue of the *Berichte der Bunsen-Gesellschaft*, some articles

apply this method, see e.g. [26, 27]. It is well known that CVTST cannot be rigorous for simple bond fission processes [20, 28–33]; e.g., thermal averaging of μ VTST does not lead to the results of CVTST. As VTST is not a dynamical theory but rather based on an intuitive minimum flux criterion, for each potential its accuracy has to be validated by comparison with dynamical theories such as rigorous SACM or CT. The situation improves when *E*- and *J*-specific μ VTST is used although the effort then approaches that of the more rigorous SACM/CT [33]. It appears improbable that the complicated avoided crossing pattern of adiabatic channel potential curves, e.g. in linear rotor + linear rotor systems [22, 23], can be simulated by μ VTST sufficiently well. For this reason, the successive calculation of specific and thermal rigidity factors for increasingly complex potential situations by SACM/CT techniques may provide a more convenient and more reliable approach to the kinetic quantities.

2. Rigidity Factors for Ion-Dipole Potentials with Uniform and Non-Uniform Anisotropies

The starting point of each determination of adiabatic channel potential curves $V_i(r)$ is the potential energy surface. For illustration, first, we consider the simple ion-permanent dipole potential

$$V(r, \gamma) = -q\mu_D \cos \gamma / r^2 + l(l+1)\hbar^2/2\mu r^2 \quad (2.1)$$

(r = center of mass distance between the point charge and the linear dipole rotor, γ = angle between the rotor axis and the line connecting the centers of mass, μ = reduced mass, q = ionic charge, μ_D = dipole moment, and l = quantum number of orbital rotation). For sufficiently large l , one may approximate l by the quantum number J for rotation of the ion-dipole adduct. Considering only the maximum attraction of the potential (2.1) ($\cos \gamma$ replaced by unity, “locked-dipole case”, PST), adiabatic channel potential curves $V_i(r)$ are given by the sum of Eq. (2.1) and rotational quanta $Bj(j+1)$ of the two-dimensional dipole rotor; channel maxima are located at $r \rightarrow \infty$ and have the energies $Bj(j+1)$ (B = rotational constant of the dipole rotor). The corresponding number of open channels is a step function which approaches the classical expression

$$W^{\text{PST}}(E, J) = E/B \quad (2.2)$$

Eq. (2.2) is only valid as long as

$$0 \leq G \leq 1 \quad (2.3)$$

with

$$G = J(J+1)\hbar^2/2\mu q\mu_D; \quad (2.4)$$

otherwise, $W^{\text{PST}}(E, J) = 0$. Replacing the locked dipole by a dipole moving in the potential of Eq. (2.1), generates rigidity with $W < W^{\text{PST}}$. Fig. 1 shows the real $W(E, J)$ in

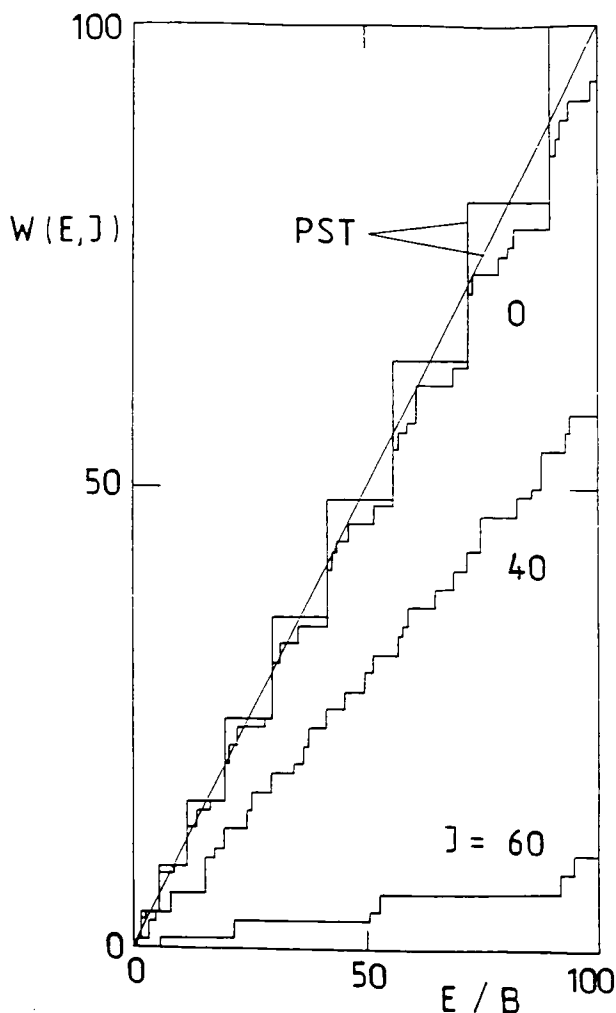


Fig. 1
Number of open channels $W(E, J)$ for the HCl-H_3^+ system (SACM: lower step functions for $J = 0, 40$, and 60 ; from [15])

comparison to the step function of $W^{\text{PST}}(E, J)$ which, in smoothed form, is given by Eq. (2.2). Fig. 1 corresponds to an H_3^+-HCl ion pair (for which $2\mu q\mu_D/\hbar^2 = 4442$ and $J_{\text{max}} = 66$). Smoothing all curves in Fig. 1 leads to the important conclusion that, for the ion-permanent dipole system, specific rigidity factors $f_{\text{rigid}}(E, J)$ are J -dependent but nearly energy independent.

Fig. 2 shows the energy independent rigidity factors as a function of G for a variety of ion-dipole systems [15]. All points are on one curve which can well be approximated by

$$f_{\text{rigid}}(E, J) = 1 - G - 0.064 \exp(-9G) \quad (2.5)$$

or, less accurately, by

$$f_{\text{rigid}}(E, J) \approx 1 - G \quad (2.6)$$

The simplicity of the classical expression (2.6) allows for a direct determination of the thermal rigidity factor $f_{\text{rigid}}(T)$ in the classical range. Inserting Eqs. (1.2), (2.2), and (2.6) into Eqs. (1.3) and (1.4) gives

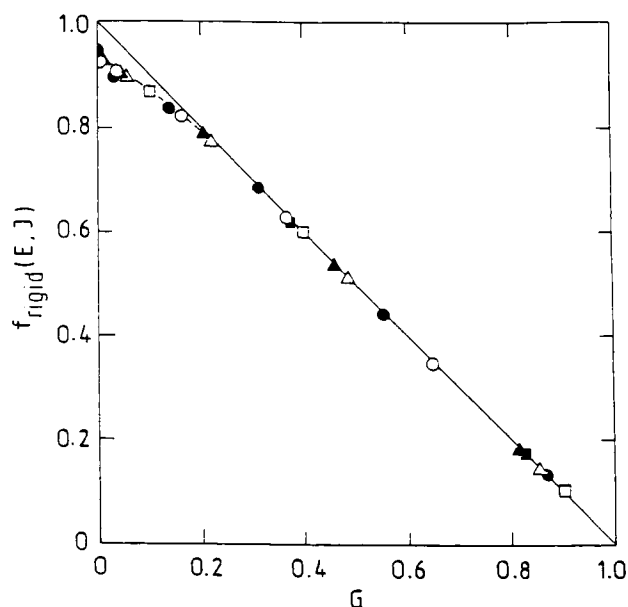


Fig. 2
Specific rigidity factors $f_{\text{rigid}}(E, J) = f_{\text{rigid}}(G)$ with $G = J(J+1)\hbar^2/2\mu q\mu_D$ for various ion-permanent dipole systems (full line: Eq. (2.6); dashed line: Eq. (2.5); from [15])

$$f_{\text{rigid}}(T) \approx \int_0^1 (1-G) dG / \int_0^1 dG = 1/2 \quad (2.7)$$

This simple result agrees perfectly with CT calculations of capture rates between ions and permanent dipoles [6, 21].

The potential of Eq. (2.1) exhibits "uniform anisotropy character", i.e. it presents the same $\cos \gamma$ -anisotropy at all distances r . In reality, the ion-dipole potential will also have other components of anisotropy. E.g., for the ion-permanent + induced dipole system, Eq. (2.1) is replaced by

$$V(r, \gamma) = -q\mu_D \cos \gamma / r^2 - \alpha q^2 / 2r^4 + l(l+1)\hbar^2 / 2\mu r^2 \quad (2.7)$$

with the dipole polarizability α . An induced dipole will make the potential relatively more isotropic at shorter than at longer distances. Specific rigidity factors $f_{\text{rigid}}(E, J)$ have been calculated for ion-permanent + induced dipole systems, again by SACM/CT techniques, in [15]. Fig. 3 shows an example, the H_3^+-HCN system with artificially varied polarizabilities. Now, the $W(E, J)$ curves are curved upward, because channel maxima move to smaller r with increasing energy where the potential is less anisotropic [21]. Also there are J -dependent threshold energies $E_0(J)$ for dissociation/recombination which can be approximated analytically in terms of the parameters

$$C = \alpha B / 2\mu_D \quad (2.8)$$

and G such as defined in Eq. (2.4) [15].

As a consequence of the curved plots of the smoothed $W(E, J)$ from Fig. 3, specific rigidity factors now are functions of J and E . It was shown in [15] that analytical approximations of $f_{\text{rigid}}(E, J)$ can be expressed as a function

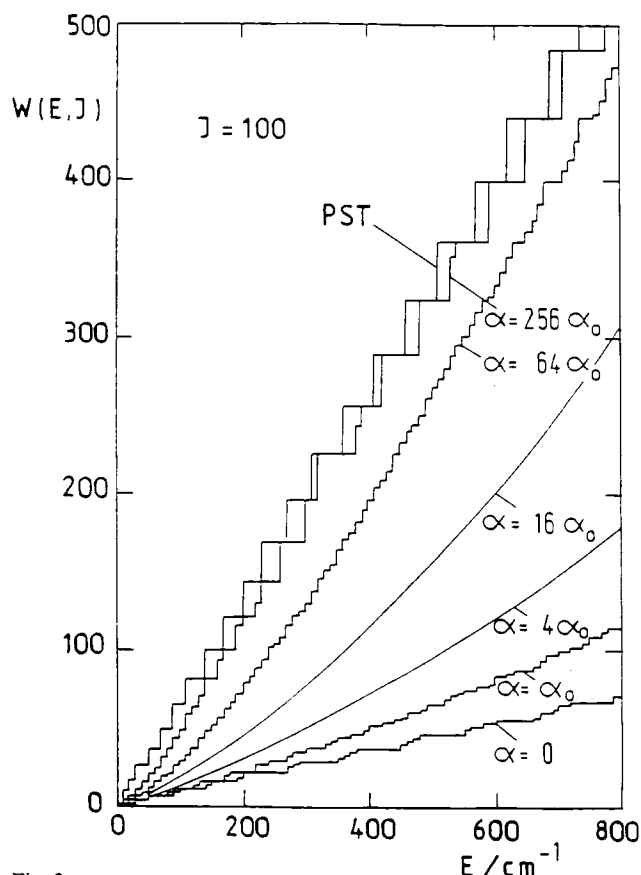


Fig. 3
Number of open channels $W(E, J)$ for the HCN-H_3^+ system with arbitrarily varied polarizability (α_0 = real polarizability, $J = 100$; PST with $\alpha = \alpha_0$; SACM with varied α , curves for $4\alpha_0$ and $16\alpha_0$ smoothed; from [15])

of the two parameters C and G . Denoting the rigidity factor of the ion-permanent dipole system from Eq. (2.5) by f_{rigid}^0 and $E^* = E - E_0(J)$, the relation between $f_{\text{rigid}}(E, J)$ and $f_{\text{rigid}}^0(E, J)$ with

$$Y = [1 - f_{\text{rigid}}(E, J)] / [1 - f_{\text{rigid}}^0(E, J)] \quad (2.9)$$

and

$$X = E^* C / BG \quad (2.10)$$

is given by

$$Y \approx \exp(-1.31 X) \quad (2.11)$$

By inserting the specific rigidity factors into the activated complex partition function $Q^*(T)$, one may again derive the thermal rigidity factor $f_{\text{rigid}}(T)$ which now will be between 1/2 from Eq. (2.7) and unity for the isotropic ion-induced dipole system. As thermal capture rate constants for ion-permanent + induced dipole collision have been derived explicitly, in [6] (with about 0.2% precision in the classical range), the thermal averaging over specific rigidity factors then can be controlled easily by the direct result. Using the analytical expressions for the locked-dipole (PST) rate

constant and for the real capture rate constant from the CT calculations of [6], $f_{\text{rigid}}(T)$ is then obtained as a function of the parameters $\theta = \alpha kT / 2\mu_D^2$. Fig. 4 from [34] in this issue illustrates the result.

The ion-permanent + induced dipole system presented the first example of a potential of "non-uniform" anisotropy where the isotropic induced-dipole contribution at shorter distances produces a relatively smaller anisotropy than the anisotropic permanent-dipole contribution at larger distances.

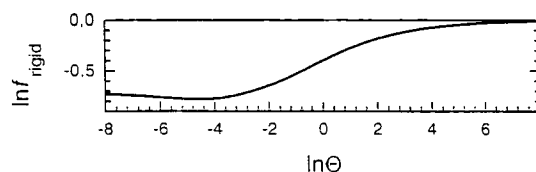


Fig. 4
Thermal rigidity factor f_{rigid} for ion-dipole capture (with $\theta = \alpha kT / 2\mu_D^2$; CT and SACM from [6])

3. Rigidity Factors for AB-C Potentials with Uniform Standard Valence Anisotropy

The SACM treatment of the ion-permanent dipole system of [21], with its analytical approximation of adiabatic channel eigenvalues from [35], can easily be adapted to valence potentials with the corresponding angular dependence. Likewise, the CT treatment of [6] can easily be extended to this case. This work has been done in [22]. In the following we demonstrate the results.

We consider a *standard valence potential* of the form

$$\begin{aligned} V(r, \gamma) = & -2D \exp[-\beta(r-r_e)] + D \exp[-2\beta(r-r_e)] \\ & + C[(1 - \cos \gamma)/2] \exp[-\beta(r-r_e)] \\ & + l(l+1)\hbar^2/2\mu r^2, \end{aligned} \quad (3.1)$$

where r is the distance between the centers of mass of a linear rotor AB and an atom C in the process $\text{ABC} \rightleftharpoons \text{AB} + \text{C}$. γ denotes the angle between the line connecting the centers of mass and the molecular axis of AB; r_e is the equilibrium value of r in the adduct ABC, β the Morse parameter and D the Morse dissociation energy. The maximum attraction part of $V(r, \gamma)$ is the standard Morse potential. The quantity $C \exp[-\beta(r-r_e)]$ is understood as the *amplitude of the anisotropy* which is assumed to be linked to the attraction of the potential in the given way by Eq. (3.1). The origin of the anisotropy part of Eq. (3.1) is the simplified SACM analysis of a series of radical-radical recombination reactions (corresponding to $\alpha/\beta \approx 0.5$ in [16]) and quantum-chemical calculations such as described, e.g. in [36] in this issue, which show that ABC bending quanta with increasing AB-C distance decrease like

$$\varepsilon(r) \approx \varepsilon(r_e) \exp[-\beta(r-r_e)/2] \quad (3.2)$$

The parameter C in Eq. (3.1) is linked to the bending quantum $\varepsilon(r_e)$ at $r = r_e$ and to the rotational constant B (in energy units) of the linear rotor AB approximately by

$$C \approx [\varepsilon(r_e)]^2/B. \quad (3.3)$$

Obviously, valence potentials in reality will be more complicated than given by Eq. (3.1). Examples are shown, e.g.; in [36] of this issue. However, the $\cos \gamma$ -anisotropy and the exponential decrease of the anisotropy amplitude provide a first approximation and, to our experience, describe already an essential part of the rigidity factors independent of the details of the PST part of the rate coefficients. Eq. (3.1) represents a case of uniform anisotropy character, against

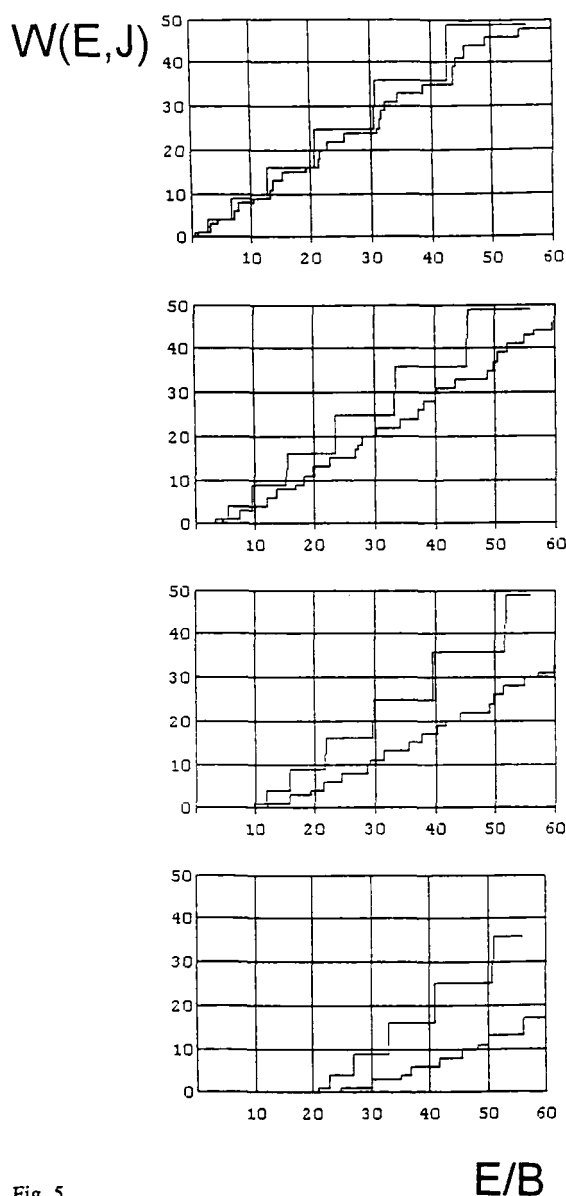


Fig. 5
Number of open channels $W(E, J)$ for $ABC \rightleftharpoons AB + C$ in a standard valence potential ($C/D = [\varepsilon(r_e)]^2/BD = 4$, see Eq. (3.3); $(C/B) \exp(\beta r_e) = 27300$; diagrams from top to bottom: angular momentum = 10, 20, 30, 40 in units of $\beta^{-1}(2\mu B)^{1/2}$; upper curves: PST; lower curves: SACM; from [22])

which non-uniformity effects can be identified by analogy to the ion-permanent + induced dipole system described in Sect. 2. Calculating adiabatic channel maxima and counting $W(E, J)$ leads to pictures such as shown in Fig. 5 which looks similar as Fig. 1 for the ion-permanent dipole case. There are again J -dependent but E -independent specific rigidity factors $f_{\text{rigid}}(E, J)$. The results can be parameterized in simple form as a function of the parameters βr_e , B , C , and D . For quantitative details the reader is referred to [22].

Thermal averaging of $f_{\text{rigid}}(E, J)$ or direct CT calculations for capture in the potential of Eq. (3.1) lead to particularly simple thermal rigidity factors. There is a weak dependence of $f_{\text{rigid}}(T)$ on T for high T and small βr_e , but in most cases f_{rigid} becomes nearly temperature independent. For this range, Fig. 6 from [22] shows the results. As long as $f_{\text{rigid}} < 0.8$, Fig. 6 can very well be represented by the simple expression

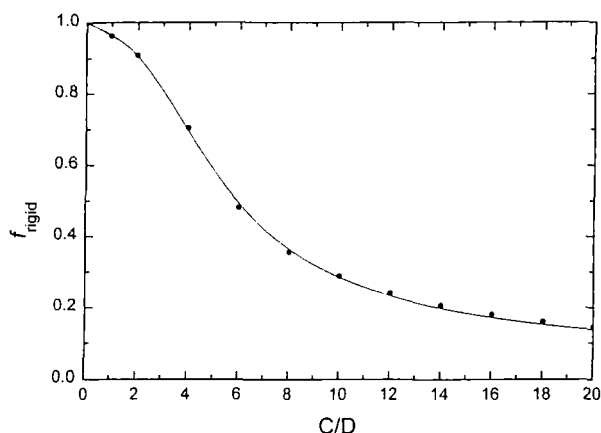


Fig. 6
Thermal rigidity factors f_{rigid} for $ABC \rightleftharpoons AB + C$ in a standard valence potential, Eq. (3.1) with $C/D = [\varepsilon(r_e)]^2/BD$ (see Eq. (3.3); classical trajectory calculations from [22])

$$f_{\text{rigid}}(T) \approx 2.85 BD/[\varepsilon(r_e)]^2 \quad (3.4)$$

(in close agreement with the crude first result from [17] where 2 instead of 2.85 was obtained).

The analysis of experimental results for a series of practical systems in [17] gave similar results as Eq. (3.4). Although each reaction will have its own special potential energy surface, it appears that at least trends of the anisotropy-governed thermal rigidity factors can quickly be estimated using Eq. (3.4).

4. Rigidity Factors for Ion-Quadrupole Potentials with Uniform Anisotropy

Our SACM/CT calculations for ion-quadrupole capture in [7] provided analytical approximations to adiabatic channel potential curves in a potential with $\cos^2 \gamma$ -anisotropy. By analogy to the transition from Sect. 2 to Sect. 3, the results could as well be adapted to valence potentials of $\cos^2 \gamma$ -anisotropy. One might determine specific and

thermal rigidity factors. Here we only note the thermal rigidity factors which are $f_{\text{rigid}}(T) = 0.385$ for $qQ < 0$ and $f_{\text{rigid}}(T) = 0.534$ for $qQ > 0$ where Q is the quadrupole moment of the linear rotor.

5. Rigidity Factors for AB-CD Potentials with Uniform Dipole-Dipole or Standard Valence Anisotropies

The systems of Sects. 2–4 had two transitional modes given by the doubly degenerate bending vibration of the linear ABC species. We now consider systems with four transitional modes. The simplest of these cases corresponds to the dipole-dipole potential treated by SACM/CT methods in [8]. Without considering specific rigidity factors, we here only note that the again temperature-independent thermal rigidity factor has the value $f_{\text{rigid}}(T) = 0.354$. If an isotropic dispersion term is added to the anisotropic dipole-dipole term, the rigidity factor increases with increasing temperature quite similar to the ion-permanent + induced dipole case of Sect. 2.

In the following, instead of the dipole-dipole case, we consider an AB-CD system with a standard valence potential. In this case, we replace the anisotropy term of Eq. (3.1), i.e.

$$V_{\text{aniso}} = C[(1 - \cos \gamma)/2] \exp[-\beta(r - r_e)] \quad (5.1)$$

by

$$V_{\text{aniso}} = (C/4) [2 - 2 \cos \gamma_1 \cos \gamma_2 + \sin \gamma_1 \sin \gamma_2 \cos(\varphi_1 - \varphi_2)] \exp[-\beta(r - r_e)] \quad (5.2)$$

Examples from the SACM/CT calculations of [22] for numbers of open channels $W(E, J)$ are shown in Fig. 7. For this system, both the smoothed PST and SACM results for $W(E, J)$ are now curved in contrast to the linear plots in Sects. 2 and 3. In deriving specific rigidity factors, of course, the small shift of the threshold energies $E_0(J)$ between PST and SACM should be accounted for. Apart from this shift, again nearly E -independent but J -dependent rigidity factors arise. For quantitative results on specific rigidity factors, the reader again is referred to [22].

The thermal rigidity factors, like for ABC systems in the potential of Eq. (5.2), for ABCD systems with uniform anisotropy are nearly temperature independent. Fig. 8 shows the results. There is a simple curve for $f_{\text{rigid}}(T)$ which, for $f_{\text{rigid}} < 0.8$, can be approximated by

$$f_{\text{rigid}} \approx [2.85 B' D / [\varepsilon'(r_e)]^2]^{1/2} \quad (5.3)$$

i.e. by the square of Eq. (3.4). One should note, however, that $B' = (B_1 + B_2)/2$ and $\varepsilon'(r_e) = [\varepsilon_s(r_e) \varepsilon_{\text{as}}(r_e)]^{1/2}$ where B_1 and B_2 are the rotational constants of AB and CD and $\varepsilon_s(r_e)$ and $\varepsilon_{\text{as}}(r_e)$ are the symmetric and asymmetric bending quanta of the linear ABCD adduct, respectively. Although real valence potential in general, of course, will be more complicated, Eq. (5.3) seems to reproduce experimental trends fairly well [17].

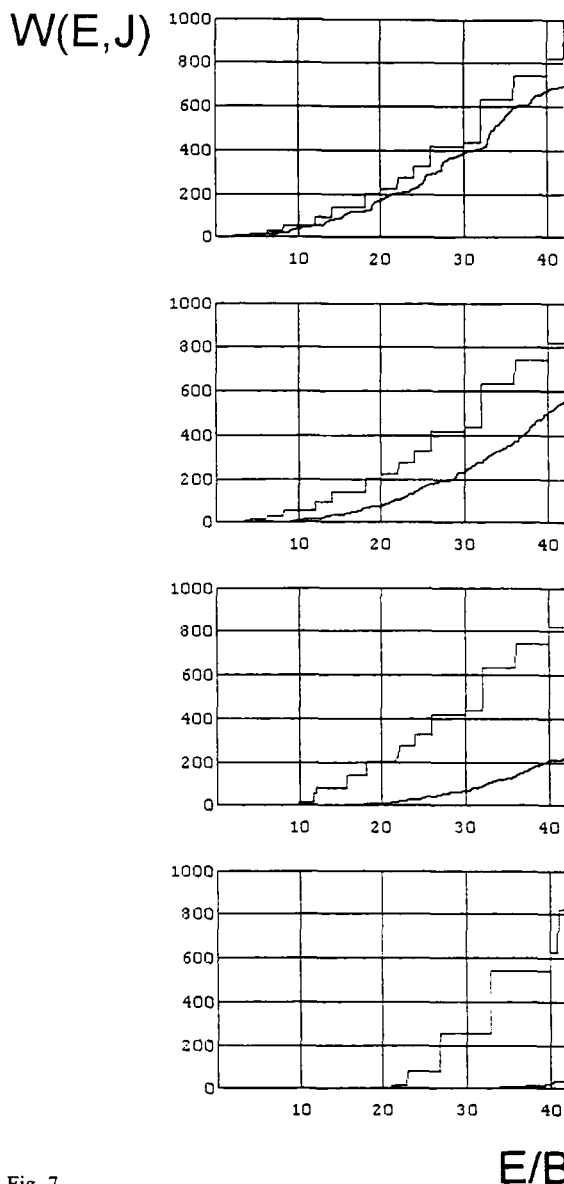


Fig. 7

Number of open channels $W(E, J)$ for $ABCD \rightleftharpoons AB + CD$ in a standard valence potential ($C/D = [\varepsilon'(r_e)]^2/B'D = 4$, see Eq. (5.3); $(C/B') \exp(\beta r_e) = 27300$; diagrams from top to bottom: angular momentum = 10, 20, 30, 40 in units of $\beta^{-1}(2\mu B')^{1/2}$; upper curves: PST; lower curves: SACM; from [22])

6. Conclusions

The discussed specific and thermal rigidity factors suggest general trends and relationships between these quantities and molecular parameters. Because thermal rigidity factors seem to vary only over two to three orders of magnitude [17], it appears reasonable to determine PST rate constants first and then add the rigidity factors. The SACM/CT techniques developed in our approach can be adapted to any attraction potential and anisotropy as long as the angular dependence is of simple form such as described in Sects. 2–5.

It should be noted that thermal rigidity factors can be used for practical comparisons such as they have been given in this article. Specific rigidity factors for two or four tran-

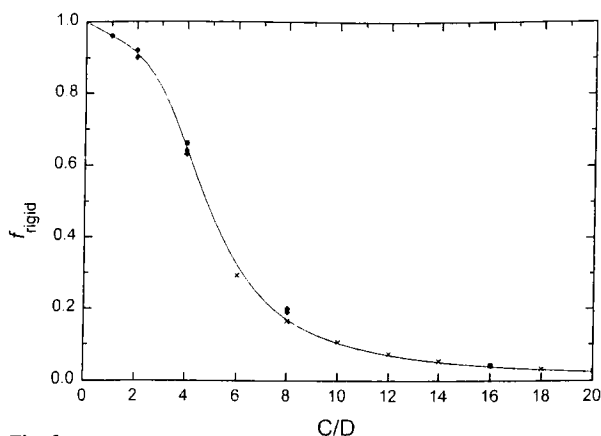


Fig. 8

Thermal rigidity factors f_{rigid} for $\text{ABCD} \rightleftharpoons \text{AB} + \text{CD}$ in a standard valence potential, Eqs. (3.1) and (5.2), with $C/D = [\varepsilon'(r_e)]^2/B'D$ (see Eq. (5.3); ● = classical trajectory calculations and x = modified harmonic oscillator model from [22])

sitional modes from this work and [22] have to be introduced into convolutions with other modes. This procedure, however, is not required when $f_{\text{rigid}}(E, J)$ does not depend on the energy.

It should be finally emphasized that real potential energy surfaces are known to be generally more complicated than our *standard valence potentials*. However, the present model calculations then still can serve as a reference point. E.g., one has often observed specific rate constants $k(E, J)$, which are close to the predictions from PST near to the threshold energy $E_0(J)$ and fall below these predictions at higher energies. This does not mean that “effective activated complexes” move to smaller distances with increasing energy, which is the general result from SACM and other approaches in any case; instead, it suggests non-uniform anisotropy with comparably stronger anisotropy character at shorter than at longer distances. This is just the opposite of what was discussed in Sect. 2 for ion-permanent + induced dipole systems and seems to be a typical property of valence potentials.

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